TASK F : Handwritten Back Propagation

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Setup, Forward Pass, Chain Rule.

Given, $x = 1$, $y = 2$ (target), $w = 0.5$, $b = 0$, $v = 1$, $c = 0$, learning rate $\eta = 0.1$

Model: $z = wx + b$, $h = \tanh(z)$, $\hat{y} = v \cdot h + c$, $J = \frac{1}{2}(\hat{y} - y)^2$

Forward Pass:

$z = (0.5)(1) + 0 = 0.5$

$h = \tanh(0.5) = 0.46211716$

$\hat{y} = (1)(0.46211716) + 0 = 0.46211716$

$e = \hat{y} - y = 0.46211716 - 2 = -1.53788284$

Local Derivatives:

$\partial J / \partial \hat{y} = (\hat{y} - y) = e$

$\partial \hat{y} / \partial e = 1$

$\partial \hat{y} / \partial v = h$

$\partial \hat{y} / \partial h = v$

$dh/dz = 1 - \tanh^2(z) = 1 - h^2 = 1 - (0.46211716)^2 = 0.78644773$

$\partial z / \partial w = x = 1$

$\partial z / \partial b = 1$

Chain Rule:

$\partial J / \partial c = e \cdot 1 = -1.53788284$

$\partial J / \partial v = e \cdot h = (-1.53788284)(0.46211716) = -0.71068205$

$\partial J / \partial w = e \cdot v \cdot (1 - h^2) \cdot x = (-1.53788284)(1)(0.78644773)(1) = -1.20946448$

$\partial J / \partial b = e \cdot v \cdot (1 - h^2) \cdot 1 = -1.20946448$

One simultaneous update, then the check & explanation

Gradient descent: $\theta_{new} = \theta - \eta \cdot \partial J/\partial \theta$, using gradients from the original parameters

$w' = 0.5 - 0.1(-1.20946448) = 0.62094645$
$b' = 0 - 0.1(-1.20946448) = 0.12094645$
$v' = 1 - 0.1(-0.71068205) = 1.07106820$
$c' = 0 - 0.1(-1.53788284) = 0.15378828$

Check with the new parameters:

$z' = 0.62094645 + 0.12094645 = 0.74189290$
$h' = \tanh(0.74189290) = 0.63028744$
$\hat{y}' = (1.07106820)(0.63028744) + 0.15378828 = 0.82886912$
$J' = \frac{1}{2}(0.82886912 - 2)^2 = 0.68577377 < 1.18254182$

$\therefore$ the loss decreased

Gradient direction: The prediction $\hat{y} = 0.462$ is below the target $y = 2$, so the error e is negative. Every gradient is therefore negative, which means increasing each parameter would increase $\hat{y}$ & reduce J. Gradient descent moves opposite to the gradient, so all four parameters increase. c changes the most because its gradient is e itself. w & b change less because the $\tanh$ slope (0.786) scales their gradients down.

C++ check (summary): central finite differences with $\varepsilon = 10^{-6}$ agree with these gradients to within about 10^{-10} (requirement $\leq 10^{-5}$)

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